

Lec 1. . Introduction

- 1st order linear system.

Sec 1

Sec 2-1, 2-2 BDM



I Introduction.

Newton's 2nd Law
 $F = ma$.

→ $x(t), v(t), a(t)$.

• Motivation.

• Advanced topics in Math
Dynamical System, PDEs.

• Applications:

gradient descent.

$$\dot{x}(t) = -\nabla f(x(t)).$$

• Mathematical modeling

• ODE: only 1 derivative

PDE: At least 2 different derivatives.

II Basic concept:

• order of ODE: unknown, $u(t)$.

n -th order:

$$F(t, u(t), \dots, u^{(n)}(t)) = 0 \quad (*)$$

e.g. $u^{(3)}(t) + [u'(t)]^2 + C = 0.$

• Linear ODE: (*) is a linear ODE if:

F is a linear function of

$$u(t), u'(t), \dots, u^{(n)}(t).$$

E.g. $u'(t) + u(t) = 0.$

• otherwise, (*) is nonlinear.

Goal of Part I:

Solve (*) explicitly for several types of ODE

III:

• First order ODE:

$$\frac{dy}{dt} = f(t, y) \quad (\text{ODE1})$$

• Simplest case.

$$\frac{dy}{dt} = C_1, \quad C_1 \text{ constant.} \quad (1)$$

$$\Rightarrow y(t) = C_1 t + C_0. \quad (2)$$

(2) contains all possible solutions to (1)

• We call $y(t)$ general solution to (1)

★: $y(t)$ is not unique due to C_0 .

• Initial value: To determine C_0 , we need additional conditions.

E.g. If

$$y(t_0) = y_0.$$

(3)

then $\Rightarrow C_0 = y_0 - C_1 t_0$

(3) is called initial condition

(1) + (3) : Initial value problem.

General method:

Reduce (ODE1) to

$$\frac{d}{dt} y(t) = g(t) \quad \text{(ODE2)}$$

$\uparrow$ unknown $\downarrow$ known

$$\Rightarrow y(t) = C + \int g(t) dt. \quad \rightarrow \text{Indefinite integral}$$

In the next few lectures, learn a few tricks to reduce the ODE to (ODE2).

IV: First order linear ODE.

Notation $y'(t) = \frac{d}{dt} y$.

$$\frac{d}{dt} y(t) + p(t)y = g(t). \quad \text{(ODE3)}$$

Goal:

ODE3 $\Rightarrow$ ODE2.

• E.g. $\frac{d}{dt} y(t) = a(t) y(t)$. put to unknown
when $y(t) \neq 0$ in $\frac{d}{dt} [\cdot]$.

observe If $y(t) \neq 0$

$$\Leftrightarrow \frac{\frac{d}{dt} y(t)}{y(t)} = a(t)$$

$$\Leftrightarrow \frac{d}{dt} \log |y(t)| = a(t)$$

$$\Rightarrow \log |y(t)| = C + \int a(t) dt.$$

$$\Rightarrow y(t) = \exp. \left[\int a(t) dt \right] \cdot C_1$$

C_1 can be any constant.

In general $\forall \mu(t) \neq 0$ ODEs $\Leftrightarrow$
 $f(t) \neq 0$
 ①
$$\mu(t) y'(t) + \underbrace{\mu p y(t)}_{\text{make it } []'} = \mu(t) f(t)$$

Find one $\mu(t)$ s.t

②
$$(\mu y)'(t) = \mu y' + \mu p y(t)$$

$$\Rightarrow \mu' y + \mu y' = \mu y' + \mu p y$$

$$\Rightarrow \mu' y = \mu p y$$

[cancel y].

$$\Leftrightarrow \underline{\mu' = \mu p.} \quad \rightarrow \text{Q: how to solve this}$$

This has been solved!!

Take one soln: $\mu(t) = \exp\left(\int_{t_0}^t p(s) ds\right)$ ③
 $\hookrightarrow$ any t_0

① + ② $\Rightarrow \frac{d}{dt} [\mu y(t)] = \mu y' + \mu p y = \mu(t) f(t)$

$$\Rightarrow (\mu y)(t) = C + \int \mu(t) f(t)$$

Given initial data

$$y(t_0) = y_0$$

obtain:

$$\mu y(t) = \mu(t_0) y(t_0) + \int_{t_0}^t \mu(s) f(s) ds$$

$$\Rightarrow y(t) = \mu(t)^{-1} y_0 + \mu(t)^{-1} \int_{t_0}^t \mu(s) f(s) ds.$$

V: Separation of variables.

Recall General form. of 1st order ODE

$$\frac{dy}{dx} = f(x, y)$$

• Rewrite.

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (\text{ODE-4})$$

$$\text{Eg } M = -f(x, y), \quad N = 1.$$

• Separable. if

$$M(x, y) = M(x), \quad N(x, y) = N(y).$$

(ODE-4) $\Leftrightarrow$

$$M(x) + N(y) \frac{dy}{dx} = 0$$

$\Leftrightarrow$

$$M(x) dx + N(y) dy = 0$$

Formally integrating $x-y \Rightarrow$

$$\int M(x,y) dx + \int N(y) dy = 0 \quad (1)$$

Indefinite
integral

Solving (1) $\Rightarrow y$.

Ex.

$$\frac{dy}{dx} e^x + e^y = 0$$

$$(1) \quad \frac{dy}{dx} e^{-y} + e^x = 0$$

$$(2) \quad e^x dx + e^{-y} dy = 0$$

$$\Rightarrow \int e^x dx + \int e^{-y} dy = 0$$

$$\Rightarrow e^x + e^{-y} + C = 0, \quad C < 0$$

$$\Rightarrow e^{-y} = -[e^x + C]$$

$$y = -\log[-(C + e^x)]$$